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Hobbesian Sovereign in 21st Century Ukraine: Assessing the Impact of Russo-Ukrainian War on Public Opinion of Volodymyr Zelenskyy

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Abstract

One of the central aspects of Thomas Hobbes's political philosophical theory is that individuals in the state of nature – a condition of war characterised by fear and uncertainty – will be inclined to grant near absolute authority and unquestionable support to a sovereign power as a means of securing protection. Establishing a parallel between this state and the conditions prevalent in modern-day Ukraine as a result of the Russo-Ukrainian war, this paper aims to test whether this aspect of Hobbes's theory holds in practice. To do so, we analyse monthly fluctuations in public opinion of Volodymyr Zelenskyy, the Ukrainian president who embodies the Hobbesian sovereign, from May 2019 to November 2023. We employ the method of sentiment analysis on a randomised sample of 7,789 Twitter (X) posts made by Ukrainians in that time frame, adopting lexicon-based classification approaches of LIWC, AFINN, and VADER. Subsequently, we use the qualitative and quantitative data to situate the detected fluctuations in sentiment against the changing political climate in Ukraine. We observe a negative status quo sentiment towards Zelesnkyy, with significant anomalous surges in positive sentiment during times of most acute fear and uncertainty in Ukrainian society, particularly in February 2022, at the beginning of the war. Consequently, we conclude that our evidence is consistent with Hobbes's theory. This provides his largely philosophical approach with an empirical backing, as well as allows us to view Hobbes's theory as offering an informative and useful lens for interpreting dynamics within warring societies.

Keywords: Hobbes, Ukraine, Zelenskyy, political theory, public opinion

Introduction

Across nearly all philosophical traditions, there exists ample literature which examines the relationship between a ruler and his subjects. This dynamic inevitably forms the basis of any political society, and thus it is unsurprising that it has perplexed the discipline's most prominent thinkers. Among them is Thomas Hobbes, an English philosopher of the late 16th and 17th centuries and a figure central to the Enlightenment tradition. His *'Leviathan'* (1651) contributes to the debate by offering a theory of a social contract between the ruler and the ruled, whereby the latter submit to the absolute power of the former in exchange for his protection. In other words, Hobbes argues that when men are in a state of war, it is natural for them to endorse a sovereign in order to escape it.

It is customary in scholarly debate to evaluate theories like that of Hobbes on a purely theoretical basis. Relative merits and shortcomings are judged in a manner that is divorced from the reality of the human condition as it presents itself in modern societies. Such an approach is expected, given that political philosophers typically build their theories relying on contexts that are hypothetically plausible, but rarely observable in practice. Indeed, Thomas Hobbes (1651: 79), who reasons from envisioning men in a state of nature, which he perceives as a state of war, acknowledges that "there had never been any time wherein particular men were in a condition of war one against another".

Nonetheless, events such as the Russo-Ukrainian war warrant bridging the gap between the theoretical approach of political philosophy and the practical side of political science. The following paper aims to do so by examining the extent to which Hobbes's theory is applicable in the real-world context of modern-day Ukraine. Fostering a bilateral relationship between the two disciplines, this paper uses the tools of the latter to assess scholarship – that is, the Hobbesian theory of the sovereign – pertaining to the former. Inversely, it draws on said scholarship to help explain major political phenomena. Research presented in this paper can be viewed as both evaluative of Hobbes's theory and instrumental in enhancing our understanding of the dynamics of the Russo-Ukrainian war. Ultimately, this paper argues that the Russo-Ukrainian war has produced outcomes that are consistent with relevant elements of Hobbes's theory, with Ukrainians being significantly more likely to support Volodymyr Zelenskyy, and to endorse his sovereign power, at the times of most acute fear and uncertainty.

In building this argument, Section 1 of this paper will begin by providing theoretical background to the research. It will explain Hobbes's account of human society and establish the integral parallels between its key pillars and modern-day Ukraine. It will show that the conditions created by the Russo-Ukrainian War closely resemble those from which Hobbes builds his theory, and that the Ukrainian president, Volodymyr Zelenskyy, can be taken as a manifestation of Hobbes's sovereign ruler. Hence, it will formulate a hypothesis based on Hobbes's theory and justify testing this hypothesis in the context of Ukraine. This section will also provide a brief overview of the existing research in the field.

Section 2 will explain the main methods that have been used to test the hypothesis. It will justify this paper's use of sentiment analysis of a large randomised sample of Tweets, and

explain how it has been used to assess the implications of the Russo-Ukrainian war for the public perception and support (or lack thereof) for Volodymyr Zelenskyy. Following thorough methodological considerations, it will outline the reasoning behind employing lexicon-based classification approaches of LIWC, AFINN, and VADER in sentiment analysis, as well as the OLS regression in further data processing.

Section 3 will present the results that the research has yielded. It will demonstrate that, against prevailing negative attitudes towards Zelenskyy, evidence shows clear points of anomalously positive sentiment. It will further show that the most consistently identified surge in support for a sovereign power appears to be during February 2022, at the beginning of the war. It will then consider and contextualise subsequent surges, suggesting that they are similarly associated with times of most acute fear and uncertainty in Ukrainian society – that is, with conditions that most closely mimic the Hobbesian state of war.

Section 4 will analyse these results to establish that the hypothesis motivated by Hobbes's theory holds, but only in the short-run. In the long-run, the support for the sovereign power diminishes and sentiment returns to the negative status quo. It will suggest that this is motivated by the reduction in the potency of conditions which characterise the Hobbesian state of war. This section will also present secondary data to supplement this paper's findings. Finally, it will examine possible limitations, as well as suggest areas for further research.

Section 1: Theoretical Background and Existing Research

Introduction

Thomas Hobbes's (1651) *Leviathan* presents a complex and intricate theory, tackling the questions concerning both the individual and the society. This paper's focus is to examine the implications of the Russo-Ukrainian war for the political power and popularity of Volodymyr Zelenskyy, as measured by the changes in the Ukrainian citizens' public opinion of him. The directly relevant aspects of Hobbes's theory, then, include his account of the state of war and the subsequent justification this provides for sovereign authority. This section first argues for strong parallels between key features of these aspects of Hobbes's account and modern-day Ukraine, before proposing a hypothesis that emerges from these.

Hobbes (1651: 77) builds his theory of the political order from theorising about the condition of men prior to their entry into civil society – that is, in the “state of nature”. On his account, the state of nature is a state of *war*, characterised by absolute freedom and, consequently, mortal danger for all human beings. His justification for such a vision is the fact that men are naturally equal in their capabilities and in competition for scarce resources. Motivated by selfish interests and in absence of a constraining authority, they can harm others for personal gain. The fundamental features of the state of nature, or war, then, are fear and uncertainty. In such conditions, Hobbes envisions “the life of man solitary, poor, nasty, brutish, and short” (ibid.: 78).

The Russo-Ukrainian war has fostered in Ukrainian society conditions which closely mimic Hobbes's state of war. The full-scale invasion of Ukraine by Russia has formally begun on the 24th of February 2022, undoubtedly seeding fear and uncertainty among Ukrainian citizens. Russian missile strikes and the presence of Russian troops across the entire country placed Ukrainians everywhere in mortal danger, akin to that which prevails in Hobbes's state of nature. Indeed, within the first month of the war, civilian casualties totaled over 1,000 persons, a number which has since been growing and reached over 10,000 two years later (Statista, 2024). Fear and uncertainty resulting from widespread and apparent lack of safety have prompted nearly 3.7 million to be displaced internally or to migrate abroad (UNHCR, 2024). Moreover, since 24th of February, Ukrainian media sources, like the reputed '*Ukrainska Pravda*' (2024) made countless reports of new attacks on civilian infrastructure – including houses, parks, and malls – therefore further intensifying the fear and uncertainty among those who have not experienced the danger first-hand. Finally, fear and uncertainty have arguably peaked at the very onset of the full-scale attack, in February 2022. The invasion began with unprecedented, for Ukrainians, deadly air strikes across the entire territory of the country, meaning that Ukrainians saw a drastic and sudden change in the status quo level of security and stability. In turn, this has been preceded by numerous intelligence reports which circulated in the media, warning citizens of the possibility of the Russian invasion in the dates which spanned across the month of February (Romaniuk et al., 2024). Overall, studies find that by August 2022, 68.2 per cent of Ukrainians surveyed showed moderate and high stress, while moderate and severe anxiety was prevalent among 25.6 per cent (Lushchak et al., 2024). Arguably, thus, central features of Hobbes's state of war were, and are, observable in modern-day Ukraine.

From his account of human existence in the conditions outlined above, Hobbes (1651: 106) reasons that a rational individual would seek protection which can be provided by a "sovereign" authority. To escape the perpetual uncertainty and the fear of premature death, men will form a social contract among themselves to authorise a sovereign as a party who can provide protection and foster peace in the society. The Hobbesian sovereign enjoys the power that is absolute. Individual subjective rights and judgments are suspended, and personal liberties are largely relinquished, in favour of the sovereign. This is motivated, first and foremost, by the prospect of security that such an arrangement can provide. Since the subjects authorise the sovereign, their personal wills are manifested in him, which effectively means that no criticism is warranted against such a figure. Individuals can neither rebel against nor judge the sovereign thus established. Therefore, Hobbes envisions a kind of exchange between the sovereign and his subjects. Motivated by fear and uncertainty, individuals forgo certain selfish desires and obey an authority to escape these dreadful conditions in order to obtain that which is most valuable – a peaceful and secure existence.

This paper views Ukrainian president Volodymyr Zelenskyy as a manifestation of a Hobbesian sovereign, as he shares certain key features with a Hobbesian ruler. Firstly, having been democratically elected, Zelenskyy implicitly represents the individual wills of Ukrainian citizens. More importantly, however, his power is visibly absolute. This is because the constitutional position of a Ukrainian president as the Head of State is such that he holds the final decision-making power on questions concerning domestic and foreign matters (The Presidential Office of Ukraine, 2024). Further, most crucially, in the conditions of the

Russo-Ukrainian war, Zelenskyy became the singular highest authority with the power to grant security and protection to his citizens against the dangers posed by the Russian invasion. The war has necessitated swift executive decisions and the use of wartime powers, and Zelenskyy has stepped up to these realities, effectively consolidating his absolute authority. To that effect, most vividly telling – for the Ukrainian public – is the fact that, with Zelenskyy's passing of the martial law in Ukraine from the very onset of the war, presidential elections have been made impossible for the duration of active combat (TVP World, 2025). Indeed, they are explicitly prohibited by Article 19 of the Law of Ukraine '*On the Legal Regime of Martial Law*' (TVP World, 2025). This has made it essentially impossible for the Ukrainian public to challenge Zelenskyy's supreme and absolute sovereignty in the Ukrainian state, making him poignantly akin to a Hobbesian sovereign.

This is where this paper's central point of inquiry comes into play. The motivating question is: What are the implications of the Russo-Ukrainian war for the public opinion of the Ukrainian president? And, to what extent does this empirical reality align with the relevant aspects of Hobbes's theory? The hypothesis which emerges following Hobbes's line of reasoning is that individuals would be less willing to challenge the authority of, or to criticise, their leader under the conditions of war – that is, at times of acute fear and uncertainty – and more inclined to grant him greater power and support. Given that these conditions have been established as prevalent in modern-day Ukraine, we hypothesise that Ukrainians would express more positive sentiment toward Volodymyr Zelenskyy following the invasion, as compared to the status quo. It is the task of the subsequent sections of this paper to test whether this hypothesis and the underlying aspects of Hobbes's theory hold in reality.

Existing Research

Much of the research on Hobbesian theory is philosophical in nature and concerns the normative implications of his arguments, rather than their consistency with the observable state of affairs in today's society. For example, a conventional line of criticism usually proclaims that a Hobbesian sovereign creates a pathway for dictatorship and authoritarianism (e.g., Onyaghola et al., 2022). Some scholars come closer to the intersection of theory with reality, examining the implications of enacting Hobbesian logic in international relations. For instance, Heller (1980: 21) responds to the argument in favour of an international sovereign, arguing that attempting to apply Hobbesian logic in practice would "raise unrealistic expectations ending in frustration". On the whole, however, academic debate does not concern itself with the descriptive elements of Hobbes's theory. The question of whether his vision of the dynamic between the ruler and the ruled holds in reality, regardless of whether we deem its consequences beneficial or problematic, is largely left unanswered, particularly with regard to the modern-day state of affairs. This, then, is the gap this paper aims to fill.

Existing research on the Russo-Ukrainian war, in contrast, spans multiple topics, analysing geopolitical effects and public sentiments. Papers predominantly fall into two categories: analyses of public sentiment or contributions to literature on sentiment classification methodology in the context of the war. For example, Guerra and Karkus (2023) study the fluctuations in feelings of hope and fear during the war, identifying spikes in these emotions

after important geopolitical events, and non-military events, like “Eurovision and football games”. Similarly, Sinha et al. (2023) use a hybrid approach to sentiment classification, employing both deep-learning and lexicon-based models to classify public opinion on multiple policy positions. In the literature examining the Russo-Ukrainian war, Chaudhri (2023) also employs sentiment analysis to gauge public opinion during the war. Moreover, Ramos and Chang (2023) examine Russian and English tweets, finding that fear and anger were predominant emotions expressed in English tweets. Multiple papers further utilise the Russo-Ukrainian conflict to test sentiment classification models, focusing primarily on methodological insights (Aslan, 2023; Nazarchuk and Albota, 2023; Mahar et al., 2023; Wadhvani et al., 2023). This paper builds on the method of sentiment analysis used by other scholars, but aims to advance the approach by considering the quantitative data thus yielded in conjunction with the qualitative analysis of the underlying political climate in Ukraine.

Section 2: Methodology

Methodological Considerations

To test the hypothesis established in Section 1, this paper analyses the nature of public opinions regarding sovereign power before and during the war. When considering methods of data collection, traditional surveys are equipped to capture attitudes at discrete points in time. However, they fall short in measuring public opinion as a continuous variable, and thus cannot account for temporal fluctuations. This is particularly relevant when considering the protracted nature of conflicts, such as the Russo-Ukrainian war. Thus, this mandates a more refined approach to analysing shifts in sentiments.

Nevertheless, this approach is not without a few challenges: first, collecting accurate public sentiments requires considerable fieldwork. Surveys are also more likely to elicit polished responses or be prone to the Hawthorne Effect, which does not reflect accurate public opinion. Second, obtaining a comprehensive understanding of the sociopolitical landscape requires a large sample that encompasses a range of political views. This would require randomisation in the participant sample, prompting further methodological refinement.

To circumvent these methodological issues while still ensuring robust analyses, this paper uses a sentiment analysis of large-scale public data from Twitter (X). Such a method offers certain advantages. First, this ensures an expansive and randomised participant sample due to the inherent properties of data scraping; an aggregation of a diverse set of user-generated content enhances the study’s validity by mitigating risks associated with selective sampling. Moreover, the platform of Twitter (X) has been validated as a medium for conducting political sentiment analysis (Kharde, 2016). Due to the unfiltered and heterogeneous composition of tweets (*ibid.*), the study of digital discourse mirrors offline emotions and political opinions, and can thus be used to study social phenomena with more power and validity.

Selecting Sentiment Classification Approaches

Popular sentiment analysis approaches involve harnessing Natural Language Processing (NLP), Machine Learning (ML) models and Deep Learning (DL) models to classify the polarity of text into positive, negative or neutral. This paper employs lexicon-based classification models for detecting complex sentiments. The decision was made to employ unsupervised models for analysis, as in addition to being highly accurate, they also override the need to manually label datasets. While attempts were made to classify and hand-label tweets for a Random Forest model and the Bidirectional Encoder Representations from Transformers (BERT) classification model, the observed accuracy was limited, and data labelling was particularly time-consuming. Thus, lexicon-based classification approaches were favoured for the final analysis.

Lexicon-based approaches are known for their speed and simplicity. By using a curated list of words associated with sentiment scores, they are ideal for establishing baseline sentiments and can be used to detect temporal shifts in opinions. Multiple papers establish the advantages LIWC confers. Ohman (2021) highlights LIWC's value in interdisciplinary social science research. More specifically, Al-Shabi (2020) identifies LIWC to be most accurate in identifying positive sentiments, outperforming deep-learning models like VADER and machine learning models like Support Vector Machines (SVMs).

Deep-learning (DL) models, in contrast, are adept in understanding linguistic subtleties. Borchers et al. (2021) identify deep learning models and SVMs to have higher precision in classifying negative sentiments, in contrast to lexicon-based approaches. The pre-trained nature of DL models streamlines implementation and analysis, reducing the time costs associated with manually labelling sentiments. However, the large computational power required for running DL models makes this particular approach challenging to implement.

Therefore, lexicon-based classification approaches to analyse Tweets presented the optimal method to fulfil the aims of this paper. The simplicity, effectiveness, and accuracy of lexicon-based approaches enable us to study both straightforward and complex sentiment expressions (Karyawati et al., 2022). Thus, the Linguistic Inquiry and Word Count (LIWC) software, AFINN, and Valence Aware Dictionary for Sentiment Reasoning (VADER) were used for data analysis. Figure 1 illustrates our methodological approach, depicting how each method addresses the needs of the research question and the ways they come together to provide evidence regarding the hypothesis.

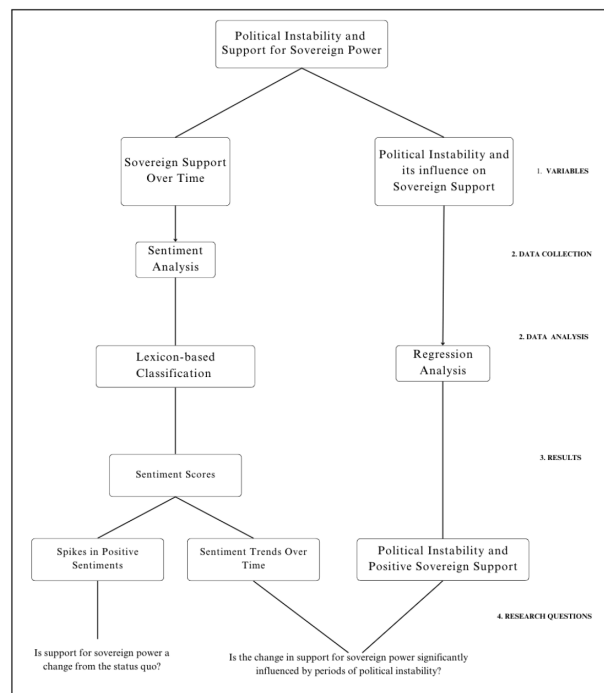


Figure 1. *Illustration of Research Methodology and Process*

Lexicon-Based Sentiment Classification

LIWC, AFINN, and VADER are lexicon-based sentiment classification methods, and the decision to employ these three approaches stems from their distinct methodologies and complementary perspectives they provide on sentiment analysis.

First, LIWC operates on a psychologically grounded lexicon, offering insights into emotional, cognitive, and structural components present in texts. This allows for deeper psychological interpretation of language; in the context of this study, analysing tweets for sentiments beyond the binary of positive and negative sentiments, such as fear, anxiety, and anger, enriches the study.

AFINN, contrastingly, is a straightforward approach that assigns fixed sentiment scores to words based on a curated and labelled dictionary of words. This simplicity and directness enable the capture of overt sentiments in texts, providing an efficient and easily interpretable analysis of polarity. In addition to its directness, AFINN is also adept at processing large volumes of data and thus serves as a useful tool for preliminary sentiment analysis.

Finally, VADER is specifically constructed to identify sentiments expressed in social media texts, making it ideal for the study's dataset. In addition to examining each individual word, VADER also considers the context of each word to determine the intensity of the sentiment. Thus, VADER proves ideal for handling the nuances and subtleties found in short-form online communication, like tweets.

Employing these three approaches creates a triangulated sentiment analysis that balances the depth of psychological interpretation with the practicality of straightforward assessment and the nuance of social media contexts. This triangulation aims to provide a more robust and comprehensive understanding of public sentiment. In a context such as the Russo-Ukrainian war, these methods capture the complexity and fluidity of political sentiments during prolonged periods of conflict.

Framework for Analysis

The questions being studied in this paper attempt to understand, at the fundamental level, the relationship between political instability and sovereign support. To do this, two crucial levels of analysis are required:

1. The fluctuations in political sentiments over periods of crisis.
2. The significance of the relationship between political instability and positive political support.

The first level is largely exploratory, focusing on understanding monthly and yearly trends in sentiments. The study first looks to understand how sentiments shift over time, before, during, and after the war. The second level attempts to quantitatively capture the effects of political instability on public sentiments. To do this, we then test whether political instability has a significant impact on sentiment trends by using measures of instability from the World Bank and the results from our primary data analysis.

With these two levels distinguished, our analysis first utilises lexicon-based approaches to classify sentiments, then progresses onto OLS regression techniques to model the relationship between positive sentiments (the dependent variable), and political instability (the independent variable), while accounting for controls.

Data

In order to ensure a randomised set of sentiments, tweets were scraped from the Twitter (X) platform. Data scraping, in our case, increases robustness while also providing a large dataset for analysis. Around 200 tweets were scraped from each month. The inclusion criteria for tweets were that they be tagged *#Zelenskyy* or variations of the name. The tweets were also restricted to have originated from within Ukraine's geographical bounds, by manually inputting the coordinates of the country into the data scraping software in such a way that the results that were yielded came from within Ukraine. Twitter (X) was chosen due to its popularity and relevance in usage during dynamic global events. Next, a randomised dataset from the entire set of scraped tweets was constructed, with 100 tweets scraped from each month. This random sampling ensures data consistency across temporal dimensions, particularly critical for unbiased machine learning models. The second advantage of random sampling is its ability to reduce noise. Randomly selecting data points mitigates the influence of outliers and anomalies, further helping the samples reflect the wider population.

Preprocessing

The scraped dataset contained 7,789 tweets, collected from the beginning of October 2019 to November 2023. Duplicates, such as retweets, were removed using the *dplyr* package. In order to sustain replication and randomisation, 100 tweets were randomly sampled from each month from the final dataset of scraped tweets to make up the final dataset for analysis. All tweets underwent preprocessing, conducted using the *tm*, *qdapRegex*, and *stopwords* packages on R. Tweets were then stemmed, and URLs, stopwords, hashtags and symbols all removed. The cleaned tweets were refined for spelling errors, and all variations of 'Zelenskyy' were corrected. The final, cleaned dataset contained 5,438 tweets ready for analysis.

Section 3: Results

This section outlines the primary analysis undertaken in this study. The first section reports on the results obtained from LIWC, followed by AFINN and VADER. The results indicate that sentiments toward Zelenskyy are generally negative over time. However, positive sentiments (such as endorsement of Zelenskyy, proclamation of his heroism, and general optimism regarding his presidency) spiked during key moments in the war. Regression results indicate a positive and significant relationship between political instability (relatedly, fear and uncertainty) and positive support for Zelenskyy, although the effect size is small. The mean difference between monthly average sentiment scores from before the war and the start of the war is statistically significant. Overall, results indicate two things:

1. While negative opinions seem to be the status quo for the Zelenskyy administration, there is, indeed, a spike in positive sentiments that are anomalous.
2. The increase in positive sentiments is significantly correlated with increasing political instability in Ukraine.

Linguistic Inquiry and Word Count

Sentiment scores

First, the LIWC software was employed to evaluate the database (LIWC, 2024). Initial examination indicated that positive sentiments comprised 35.1% of the corpus, whereas negative sentiments constituted 45.39%.

Temporal distribution of emotional sentiments was subsequently mapped. Positive sentiments peaked in February 2022 and were at their lowest in November 2021. In contrast, negative sentiments were at their highest in January 2022 and their lowest in April 2021 (Figure 2). In plotting both sentiments together, it is observed that positive sentiment scores surpass negative sentiments during November 2020, February 2022, and April 2022 (Figure 3). Emotions of 'anger' and 'sadness' exhibit slightly similar trends, peaking around early 2022, and continuing to stay heightened through 2022 and 2023 (Figure 4).

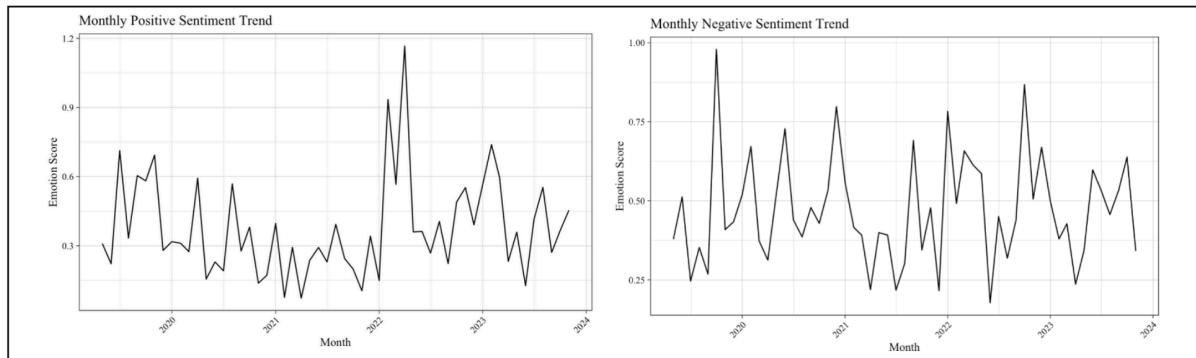


Figure 2. *Sentiment Trends from 2019-2023*

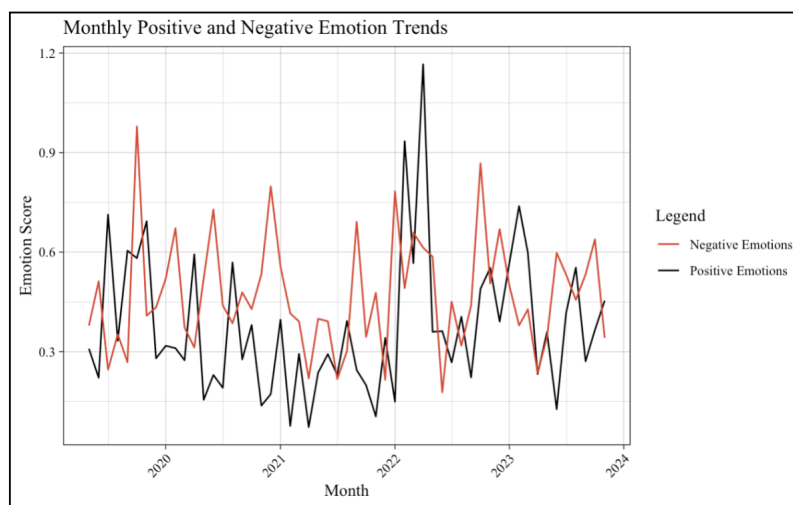


Figure 3. *Positive and Negative Sentiment Trends Over Time*

Anger and Anxiety scores are also visualised below. Anger peaked in April 2022, while Anxiety in January 2022.

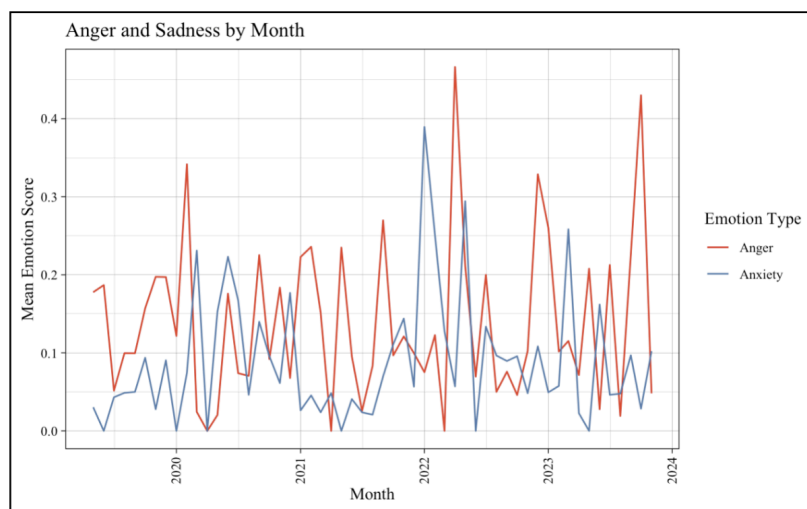


Figure 4. *Anxiety and Anger Trends Over Time*

The influence of political stability on sentiments

Next, an OLS regression was conducted to study whether political events had an influence on sentiments towards Zelenskyy. To do this, secondary data on political stability (a variable that is seen as reflecting fear and uncertainty, or lack thereof) and Gross Domestic Product (GDP) were obtained from the World Bank's World Governance Indicators, for the years 2019-2022 for Ukraine¹. Positive sentiments were regressed onto political stability first and then regressed again with GDP as an additional control. In both cases, political stability emerged as a significant predictor of positive sentiments, although the effect size is small (Table 2). The predictors explained 17.26% of the variance. The results indicated that, controlling for GDP, political stability was significantly negatively associated with positive sentiment scores ($b = -0.210$, $SE = 0.093$, $t(41) = -2.267$, $p = .029$). GDP was not significantly associated with positive sentiment scores ($b = -2.153e-12$, $SE = 1.774e-12$, $t(41) = -1.213$, $p = .232$). Therefore, we contest that greater political instability predicts greater positive support for Zelenskyy (i.e., increased political stability is associated with a decrease in positive sentiments).

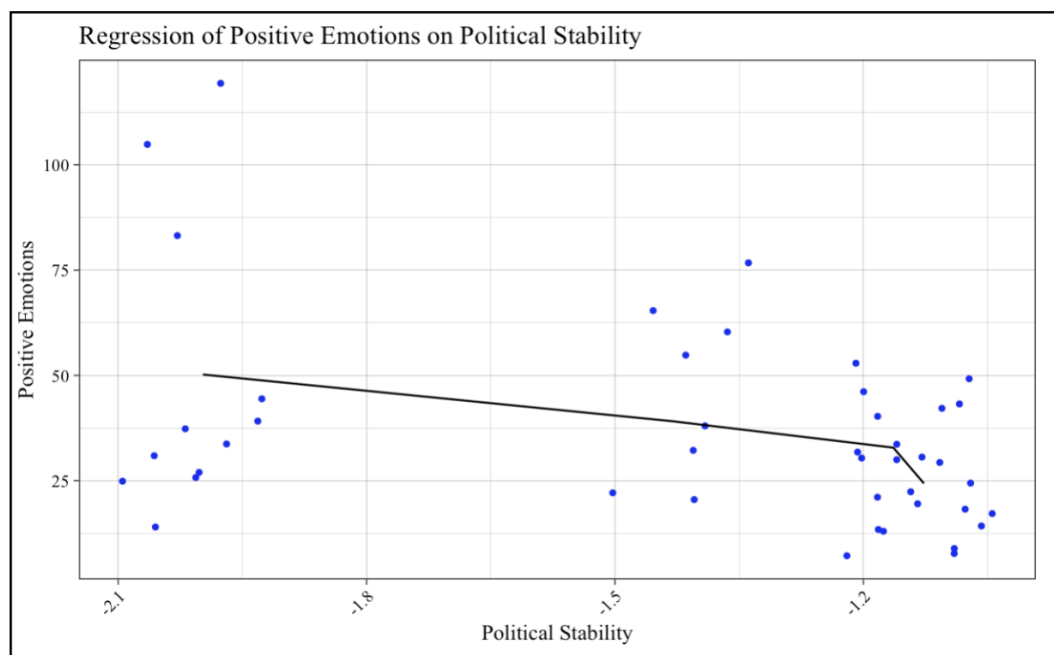


Figure 4. *Relationship between Political Stability and Positive Sentiments*

¹ The statistics for 2023 are unreleased.

<i>Variable</i>	<i>B</i>	<i>SE</i>	<i>t</i>	<i>p</i>
Positive Sentiments (Intercept)	4.287e-01	3.773e-01	1.136	0.262
Political Stability	-2.103e-01	9.274e-02	-2.267	0.029 *
GDP	-2.153e-12	1.774e-12	-1.213	0.232

Table 1. *OLS Regression Results***Statistical significance of mean sentiment score differences**

A Wilcoxon Paired Rank-Sum Test was conducted post-hoc to evaluate the differences in the distribution of positive sentiment scores across the months. The results indicated that the median positive sentiment score for February 2022 was significantly different from the median positive sentiment score of December 2021 and January 2022 ($W = 4191.5$, $p < 0.001$). This suggests that there was a statistically significant shift in the location of positive sentiment scores across the two months.

AFINN and VADER

AFINN was employed in R with the *tidytext*, *dplyr*, and *tm* packages. We classified the preprocessed corpus of tweets into positive and negative sentiments. Figure 5 below illustrates the number of tweets classified as positive and negative. Further visualising this as a time-series plot (Figure 6), we see that negative tweets predominate the time period from 2019-2023. The positive tweets peaked in 2022 and 2023. Next, we employed VADER to identify months where total positive sentiments were higher than total negative sentiments. Five such months were identified: June 2019, July 2019, February 2022, December 2022, and May 2023. Particularly, we find that in the month of February 2022, positive sentiments averaged a sentiment score of 3.08, while negative sentiments were significantly lower, at 2.57.

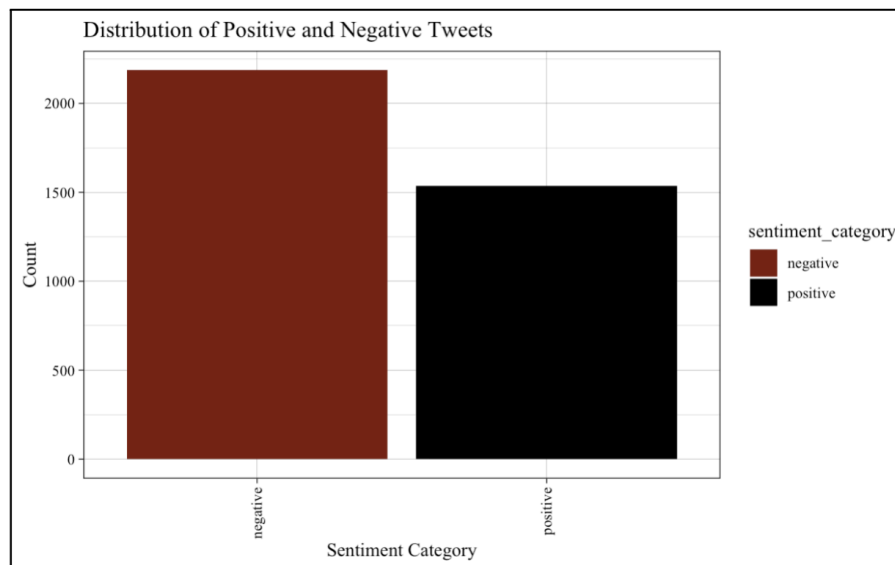


Figure 5. Overall Classification of Tweets into 'Positive' and 'Negative' (unclassified tweets were removed from the visualisation)

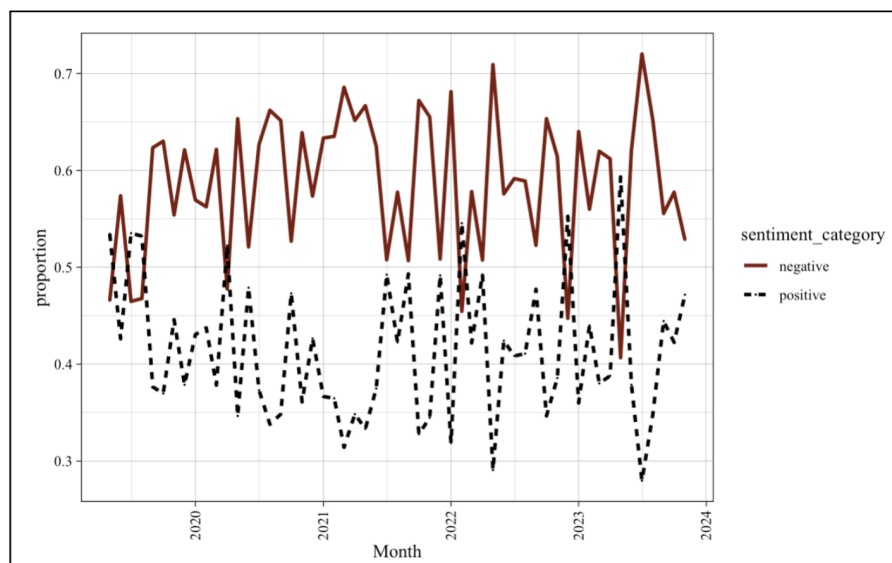


Figure 6. Proportion of Positive and Negative Tweets Over Time

Section 4: Discussion

Interpreting the Results

In discussing the results of the sentiment analysis, we will first cover the general, status quo sentiment trend observed. Subsequently, we will discuss the anomalous deviations from this trend and their relationship with political stability, parsing out their implications for our hypothesis.

The first fundamental pattern that can be observed from the data is that the status quo sentiment towards Volodymyr Zelenskyy is negative from the start of his presidency to November 2023 (the final point of data collection). It is seen most vividly in Figure 2 and in Figure 5, which shows the distribution of tweets by broadly positive and negative sentiment, with the negative exceeding by approximately 33% (Figure 5). Linguistic analysis suggests that the most frequent words used to describe him were the words “corrupt”, “nazi” or a “clown”. The lexicon-based approaches corroborate this observation, with both LIWC and AFINN demonstrating that negative sentiments exceed positive ones. This echoes existing research, for example, the finding from Gulzar et al. (2023), which suggests that most tweets on the platform are negative. These trends, in addition to being observed, are also found to have no statistically significant pairwise differences in the Wilcoxon Paired Rank-Sum Test. This firmly establishes the idea that the status quo perception of Volodymyr Zelenskyy is overwhelmingly negative.

We take this general trend to make the Ukrainian context particularly illuminating for examining Hobbes’s theory in empirical reality. This is because, against such a backdrop, the effect of fear and uncertainty, seeded by the war, can be vividly observed. Indeed, since Zelenskyy assumed office in May 2019, all methodologies employed indicate several distinct points at which Ukrainians have expressed significantly more positive than negative sentiment towards their sovereign. The LIWC approach indicates that this has been the case in November 2020, February 2022, and April 2022 (Figure 3). Meanwhile, AFINN and VADER approaches suggest that positive sentiment exceeds negative in June 2019, July 2019, February 2022, December 2022, and May 2023 (Figure 6). Finally, despite these differences, all approaches employed in this paper unanimously indicate a surge in the positive sentiment at the onset of the war, in February 2022.

What is the implication of these results for the central line of inquiry of this paper?

Most consistent evidence points towards the fact that surges of positive sentiment coincide with times of most acute fear, uncertainty and political instability in Ukrainian society. This, in turn, confirms our hypothesis and the underlying elements of Hobbes’s theory. This is most apparent in the fact that Ukrainians were more supportive than critical of Zelenskyy in February 2022. It has been previously established that this period in Ukrainian society was characterised by widespread fear of the Russian invasion and substantially increased uncertainty regarding peace and security in Ukraine. Evidently, under these conditions, Ukrainians have become more supportive of their sovereign, perhaps viewing him as a source of protection. The significant inverse relationship between political stability and positive sentiments (that is, the fact that an increase in positive sentiments is significantly correlated with increasing political instability in Ukraine) further highlights that greater fear and uncertainty tend to be associated with a more positive dialogue surrounding leadership. This effect of the state of war in Ukraine is thus consistent with one theorised by Hobbes.

Examining subsequent surges in the positive sentiment further confirms this. Firstly, regarding the less consistently identified surges, we must be careful in ascribing any significance to those that took place in June and July 2019. These months are characterised by a relatively low overall level of sentiment expression (Figure 3), likely attributed to the novelty of

Zelenskyy's persona in the Ukrainian political arena. It is reasonable to allow time to pass between the election and the point at which publicly expressed sentiment shows individuals' genuine and informed opinion of their sovereign. Subsequently, with the exception of November 2020, all surges in positive sentiment took place in wartime and correlated with events that have prompted a reduction in political stability and an increase in prevailing fear and uncertainty. For instance, May 2023 has been characterised by the largest air strikes on Kyiv and other Ukrainian territories up to that date, with over 60 kamikaze drones being fired at various targets (Reuters, 2023). Arguably, this has intensified the fear prevailing among Ukrainians. Furthermore, uncertainty has also increased due to a significant escalation of the war in that month. Moscow has been targeted with a large-scale drone attack for the first time in 15 months of conflict, and the Kremlin has threatened to take the "harshest possible measures" in response to them (Russia Post, 2023). Therefore, consistent with the hypothesis, the support for Volodymyr Zelenskyy increases while the critical attitudes diminish, when conditions characteristic of the state of war – fear and uncertainty – become more acute in Ukrainian society.

Interestingly, we find that the positive sentiment expressed by Ukrainians towards Volodymyr Zelenskyy diminishes after each surge and, in the long-run, returns to the overwhelmingly negative status quo. Indeed, following the surge in February and March 2022, the number of negative tweets exceeded the positive ones (Figure 6). One explanation for this is that the potency of conditions which motivate Ukrainians to support their sovereign – fear and uncertainty – diminishes as the war continues. In effect, faced with reality, many Ukrainians no longer hold out for the unexpected, and the society itself adapts to the state of war. Furthermore, following February 2022, the danger, arguably, has become more localised in Ukraine's eastern and south-eastern regions of Donetsk and Luhansk. Indeed, Ukraine's major counter-offensive efforts have pushed the Russian army away from its northern regions and the capital in Kyiv from as early as March 2022 (Walker, 2023). Consequently, given that the motivating factors behind increased support for a sovereign (in the context of our hypothesis) have been prevailing fear and uncertainty, the return to the negative status quo in the sentiment expressed is not necessarily inconsistent with our hypothesis.

Finally, the general trend found in this paper – that Ukrainians express an overall more positive sentiment towards Volodymyr Zelenskyy during war as compared to pre-war – can also be observed in other independent research. For example, results of computer-assisted telephone interviews conducted by a Ukrainian sociological review group indicated that 74 per cent of Ukrainians approved of Zelenskyy's actions in April 2022 (Statista Research Department's first data point since the beginning of the war), compared to only 12 per cent in March 2021 (Statista Research Department's preceding data point) (Statista, 2024). Similarly, in March 2021, nearly 28 per cent of the interviewees expressed strong disapproval of their president, while in April 2022, this number was only 2 per cent (Statista, 2024). Interestingly, this study finds that the high strong approval and low strong disapproval remain true throughout the entire wartime, until February 2023 (Statista Research Department's last available data point). Evidently, secondary evidence also confirms the findings of this paper, further supporting our hypothesis and the underlying aspects of Hobbes's theory.

Limitations and Considerations for Future Research

One limitation of this paper is that it views fluctuations in support for Volodymyr Zelensky as a function of fear and uncertainty (or, relatedly, political instability) or lack thereof in Ukrainian society. However, it may be contested that the sentiment expressed by Ukrainians depends on a set of alternative variables, such as the extent to which they approve or disapprove of the policies enacted by the president. Although this may be the case, the task of this paper has not been to ascertain the definite causation of greater support by increased fear and uncertainty, but rather to observe the correlation between these two variables. While consistent with the findings of this paper, our hypothesis and the underlying aspects of Hobbes's theory provide merely one plausible lens under which to examine the issue and need not be exhaustive. Rather, these findings must be viewed as instrumental in evaluating Hobbes's logic on an empirical basis, as well as in further understanding the dynamics of the Russo-Ukrainian war.

There are also methodological limitations that must be addressed. First, Twitter (X) is one of many social networking sites, and the user base may not accurately reflect all public sentiments. While care has been taken to randomise the data sample to ensure it is representative, future research could consider a more refined methodology, perhaps sourcing data from multiple online platforms. Second, lexicon-based approaches, while effective, do not fully consider all contextual information associated with texts, which may limit their accuracy. In light of this, future research could employ DL models equipped to study sentiments within their contexts, or ML models, like Random Forest or Naive Bayes. While considerable effort is required to label, code, and train these models, the outputs benefit from higher accuracy. While existing research on the Russo-Ukrainian war indeed employs DL models (Aslan, 2023; Wadhwani et al., 2023; Vahdat-Nejad et al., 2023), the analysis was constrained to understanding public sentiments. This study builds on existing literature by adding another dimension to the analysis by examining sentiment fluctuations alongside political instability. However, expanding the sample size and exploring additional controls in the regression model can increase validity and generalisability. Finally, while this study tests English tweets, further analysing tweets in the Ukrainian language could provide a more nuanced understanding of shifts in public sentiments.

Conclusion

This paper has focused on examining the implications of the Russo-Ukrainian war for the public opinion of the Ukrainian president, Volodymyr Zelenskyy. Following the line of reasoning adopted by Thomas Hobbes in the *'Leviathan'*, it has been hypothesised that individuals would be less willing to challenge the authority of, or criticise, their leader under the conditions of war. At times of acute fear and uncertainty (relatedly, political instability), citizens are more inclined to grant the sovereign greater power and support. This hypothesis has been tested via collecting and analysing a large randomised sample of Twitter (X) posts made by Ukrainians in the period of Zelenskyy's presidency, and subsequently using lexicon-based language classification methods of LIWC, AFINN, and VADER to detect monthly fluctuation in negative and positive sentiments. The evidence gathered strongly suggests a negative status

quo opinion of Zelenskyy, against which surges of positive sentiment (i.e., greater expressed support and reduced expressed criticism) distinctly stand out. A surge that is most consistently identified across all methods, as well as statistically significant, took place in February 2022. This coincided with the onset of the war, during the period of most acute fear and uncertainty. This evidence is consistent with our hypothesis and the underlying aspects of Hobbes's theory. Further analysis of subsequent surges in positive sentiment corroborates these conclusions, as do the findings of the OLS regression, according to which political instability predicts greater positive support for Zelenskyy. We can thus conclude that the Russo-Ukrainian war has produced outcomes that are consistent with relevant elements of Hobbes's theory, with Ukrainians being significantly more likely to support Volodymyr Zelenskyy and to endorse his sovereign power at the times of most acute fear and uncertainty.

In making this argument, this paper has aimed to foster a bilateral relationship between the disciplines of political science and political philosophy. The goal has been to evaluate some aspects of Hobbes's philosophical theory on the descriptive basis, in the tangible realm of human affairs, thereby contributing to scholarly debate which tends to focus on the normative side of this theory. Meanwhile, the decision to utilise a Hobbesian explanation for the phenomena observed in the context of the Russo-Ukrainian war aims to provide an insightful and useful lens for considering current political events, rather than claiming to be exhaustive or to ascertain a definite causal mechanism.

Naturally, we concede that the application of the 17th-century philosophical theory to the 21st-century reality, undertaken by this paper, has been far from perfect. Conditions observed in Ukraine do not exactly replicate those which Hobbes had used to build his argument. Meanwhile, the explanation for the relevant political phenomena provided by Hobbes's account becomes merely one angle from which to consider the issue, and it is plausible that said phenomena were driven by a multitude of other factors. Nonetheless, we find that the striking parallels between certain key features of Hobbes's reasoning and the conditions observable in modern-day Ukraine are sufficient to render such an inquiry informative, both for political scientists and philosophers.

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